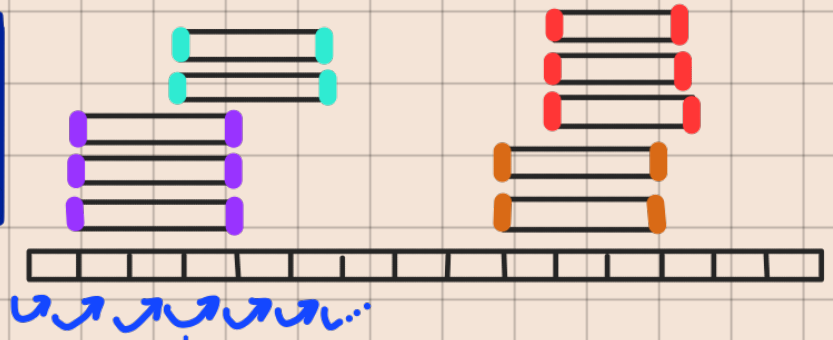
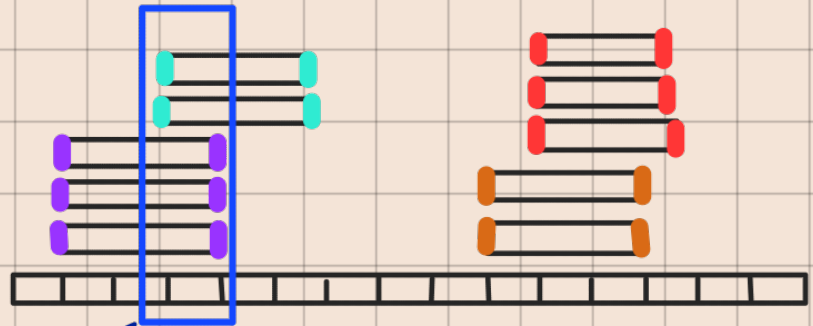


First, iterate over the loci



Next, iterate over the reads at that locus.



pileup at the current locus:

UMI 1 { A  
A  
...  
A  
UMI 2 { A  
T  
...  
A

$$\epsilon_{UMI1} = \frac{\# \text{ major alleles}}{\# \text{ reads with UMI 1}}$$

$$w_{UMI1} = \# \text{ reads with UMI 1}$$

$$\epsilon_{UMI2} = \frac{\# \text{ major alleles}}{\# \text{ reads with UMI 2}}$$

$$w_{UMI2} = \# \text{ reads with UMI 2}$$

Finally, calculate the weighted average error rate per locus.

$$\epsilon_{\text{locus } x} = \text{wmean}(\epsilon_{UMI1} \times w_{UMI1}, \epsilon_{UMI2} \times w_{UMI2})$$